

Convex Optimization: Problem Set 1

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Some problems borrowed from Optimization course of Daniel Dadush.

1. Recall the definition of convex hull:

$$\text{conv}(S) := \left\{ \sum_{i=1}^N \lambda_i \mathbf{x}_i \mid \mathbf{x}_i \in S, \lambda_i \geq 0, \sum_{i=1}^N \lambda_i = 1 \right\}.$$

1. Prove that for finite S , $\text{conv}(S) = \cap_{K \supseteq S} K$ where the intersection is over all closed convex K containing S . Therefore $\text{conv}(S)$ is the smallest convex set containing S .
2. Prove Jensen's inequality: for convex $f : \mathbb{R}^n \rightarrow \mathbb{R}$ and input $\mathbf{x} = \sum_{i=1}^N \lambda_i \mathbf{x}_i$ where $\lambda_i \geq 0, \sum_{i=1}^N \lambda_i = 1$

$$f(\mathbf{x}) \leq \sum_{i=1}^N \lambda_i f(\mathbf{x}_i).$$

3. Show that if $\mathcal{C} = \text{conv}\{x_1, \dots, x_m\} \subseteq \mathbb{R}^n$ is a compact convex set and $f : \mathbb{R}^n \rightarrow \mathbb{R}$ is a convex continuous function then the **supremum** of f over $\mathcal{C}$ is attained at one of the vertices x_j .
2. Show the following are equivalent and prove any one of them:
 - (a) Cauchy-Schwarz inequality for $\mathbb{R}^n$

$$\langle x, y \rangle = \sum_{i=1}^n x_i y_i \leq \sqrt{\sum_{i=1}^n x_i^2} \sqrt{\sum_{i=1}^n y_i^2} = \|x\|_2 \|y\|_2;$$

- (b) The triangle inequality $\|x + y\|_2 \leq \|x\|_2 + \|y\|_2$;
 - (c) Convexity of the Euclidean norm $f(x) := \|x\|_2$;
 - (d) Convexity of the unit ball $B_2^n := \{x \in \mathbb{R}^n \mid \|x\|_2 \leq 1\}$.
3. For convex closed $f : \mathbb{R}^n \rightarrow \mathbb{R}$ and $x \in \text{dom}(f)$:
 - Show the sub-level set $L := \{y \mid f(y) \leq f(x)\}$ is a convex set.

- Show for any subgradient $g \in \partial f(x)$, $(g, -1)$ gives a *supporting hyperplane* at $(x, f(x))$ for the epigraph

$$\text{epi}(f) := \{(x, t) \mid f(x) \leq t\}.$$

- Show that $g \in \partial f(x)$ gives a *supporting hyperplane* at x for the sub-level set

$$L := \{y \in \mathbb{R}^n \mid f(y) \leq f(x)\}.$$

4. Affine and quadratic functions are the most basic convex functions. We will prove some properties about them:

- Show $h : \mathbb{R}^n \rightarrow \mathbb{R}$ is convex *and* concave (i.e. h is convex and $-h$ is convex) iff h is an affine function, i.e. $h(x) = \langle a, x \rangle + b$ for $a \in \mathbb{R}^n, b \in \mathbb{R}$.
- Let $\tilde{q}(x) := \langle x, Qx \rangle + \langle a, x \rangle + b$ for symmetric matrix $Q \in \mathbb{R}^{n \times n}, a \in \mathbb{R}^n, b \in \mathbb{R}$. Show $\tilde{q}$ is convex iff $q(x) := \langle x, Qx \rangle$ is convex. (Hint: use first part)
- Show $q(x) = \langle x, Qx \rangle$ is convex iff

$$\forall v \in \mathbb{R}^n : \quad \langle v, Qv \rangle \geq 0,$$

and similarly show it is strictly convex iff the inequality is strict for any $v \neq 0$. You can use any of the equivalent characterizations of convexity.

(These conditions are known as positive-semi-definiteness and positive-definiteness, and are denoted $Q \succeq 0, Q \succ 0$.)

- Find the optimizer and optimum value of strictly convex quadratic

$$\tilde{q}(x) := \langle x, Qx \rangle + \langle a, x \rangle + b.$$

5. An Ellipsoid is an affine image of the Euclidean ball

$$\mathcal{E} = c + AB_2^n \quad \text{where} \quad B_2^n := \{x \in \mathbb{R}^n \mid \|x\|_2 \leq 1\},$$

for some $c \in \mathbb{R}^n$ and $A \in \mathbb{R}^{n \times n}$ invertible.

- Let $q(x) := \langle x, Qx \rangle + \langle d, x \rangle + e$ be *strictly convex*. Show any sub-level set

$$L_t := \{x \in \mathbb{R}^n \mid q(x) \leq t\}$$

is either empty or an Ellipsoid (i.e. find c, A such that $L_t = c + AB_2$).

- Conversely, given Ellipsoid $\mathcal{E} = c + AB_2$ as above, find convex quadratic $q(x) := \langle x, Qx \rangle + \langle d, x \rangle + e$ such that

$$\mathcal{E} = \{x \in \mathbb{R}^n \mid q(x) \leq 1\}.$$

6. Recall the GLS oracle model for convex sets. For convex set $K \subseteq \mathbb{R}^n$, the *polar dual* is

$$K^\circ := \{w \in \mathbb{R}^n \mid \sup_{x \in K} \langle w, x \rangle \leq 1\}.$$

- Relate VAL and VIOL oracles for K to MEM and SEP oracles for K° ;
- Combine this with cutting plane methods to show a reduction from SEParation to OPTimization. In more detail, given access to OPT oracle for K , show how to implement the (approximate) SEParation oracle for K .

This may seem counter-intuitive, but there are important settings where OPT is easier to implement than SEP (e.g. when K is the spanning tree polytope of a graph).